Divergence test - 5.3

Divergence test: If $\lim_{n\to\infty} a_n$ fails to exist or exists but is nonzero, then $\sum_{n=0}^{\infty} a_n$ diverges.

pf. We will prove that $\sum_{n=0}^{\infty} a_n$ converging implies that $\lim_{n\to\infty} a_n = 0$.

To say that $\sum_{n=0}^{\infty} a_n$ converges means that $\lim_{k\to\infty} s_k = L$. Then $a_n = s_n - s_{n-1}$, so

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} s_n - s_{n-1} = L - L = 0. \quad \square$$

$$\sum_{n=1}^{\infty} 1 \quad \text{diverges}$$

$$\sum_{n=1}^{\infty} \frac{n}{n+1} \quad \text{diverges}$$

$$\sum_{n=1}^{\infty} (-1)^n \quad \text{diverges}$$

Just because the limit of a_n is zero, we can't conclude that $\sum_{n=1}^{\infty} a_n$ converges.

Example: $\sum_{n=1}^{\infty} \frac{1}{n}$ has $\lim_{n \to \infty} \frac{1}{n} = 0$

but does not converge. "Harmonic series"